FICTIONAL TEST DISCLOSURE: made up for testing, not a real invention

Invention Disclosure Form

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Title of invention	Self-calibrating acoustic leak detection for district heating pipes
Inventors	Dr. C. Example (staff scientist, 60%), M.Sc. D. Example (doctoral researcher, 40%)
Department	Department of Mechanical Engineering (fictional)
Date of disclosure	1 October 2026
Funding of the research	Research Council of Finland project (2024 to 2027); an energy company provided access to a test network but no money
Planned publications	Conference paper submitted on 15 September 2026, decision expected in December 2026

1. Field of the invention

Monitoring of underground district heating networks, in particular detecting and locating water leaks in pre-insulated steel pipes.

2. Problem

District heating networks lose hot water and heat through leaks that are hard to find. Operators today rely on the alarm wires built into pre-insulated pipes, thermal drone flights, and manual listening with ground microphones. Alarm wires are broken or missing in many older networks, and drone flights only work in winter and give a rough location. Finding a leak can take days of digging.

3. Description of the invention

Small battery-powered acoustic sensors are clamped onto the pipes in existing valve chambers, typically 300 to 800 m apart. Each sensor records pipe vibration for a few minutes every night, when the network is quiet. A leak produces a broadband hiss that travels along the pipe in both directions.

The new part is that the system **calibrates itself**. Each sensor sends a short known vibration pulse into the pipe, and the neighbouring sensors measure how fast it travels and how much it weakens. From this the system learns the actual sound speed and damping of each pipe section, which depend on pipe size, insulation, age and water temperature. Existing correlators need these values entered by hand and are often wrong by 10 to 20%, which moves the estimated leak position by tens of metres. A small model on the server combines the night recordings with the learned pipe properties to flag leaks and estimate their position.

4. Results so far

Test	Result
Lab pipe rig (DN100, 40 m)	Simulated leaks of 0.5 l/min detected; position error under 1 m

Field trial, 2.3 km of network, 6 sensors, 4 months	3 known leaks found; mean position error 2.1 m (manual correlator on the same data: 9.4 m)
False alarms in the field trial	2 in 4 months, both caused by a pump station
Battery life (estimated)	About 5 years with nightly recording

The sensors in the trial were built from off-the-shelf parts; the estimated part cost is about 120 EUR per sensor.

5. Known prior art

Acoustic leak correlators are widely used for drinking water pipes, and some vendors offer fixed acoustic loggers. We are not aware of systems that measure the sound speed and damping of each pipe section automatically with active pulses. We have not done a patent search.

6. Commercial interest

The energy company that hosted the field trial has asked about a larger pilot in 2027. No other companies have been contacted.

7. Team and plans

The doctoral researcher would like to start a company after the doctorate (expected 2027). The staff scientist wants to stay in research but could advise. Nobody in the team has business experience.